

Practice 2

Lesson 2 Assessment

Variables, Input, and Build a Personal Tool

Part A. Single Choice Questions (10 pts)

1. What is a variable?

- A. A command that prints text to the screen
- B. A named storage location that holds a value
- C. A type of loop
- D. A compiler error

Answer:

2. Which statement correctly declares an integer variable in C++?

- A. `integer age = 18;`
- B. `num age = 18;`
- C. `int age = 18;`
- D. `var age = 18;`

Answer:

3. What will be stored in `message` ?

```
string name ="Alice";  
string message ="Hello "+name;
```

- A. `Alice`
- B. `Hello`
- C. `Hello Alice`
- D. Compilation error

Answer:

4. Which variable name is valid in C++?

- A. `2score`
- B. `my-score`

C. `class`

D. `playerScore`

Answer:

5. What is the output?

```
int x =10;  
x =x+5;  
cout<<x;
```

A. 5

B. 10

C. 15

D. 105

Answer:

Part B. Fill in the Blanks (10 pts)

1. A variable stores a _____ that can change during program execution.

Answer:

2. The C++ keyword used to declare a string variable is:

```
_____name;
```

Answer:

3. The operator used to combine two strings is:

```
_____
```

Answer:

4. Complete the code:

```
int age =15;  
cout<<"Age: "<<_____;
```

Answer:

5. After execution, the value of `counter` is:

```
int counter =0;
counter =counter+1;
counter =counter+1;
```

Answer:

Part C. True or False (10 pts)

1. A variable can store different values at different times.

Answer:

2. The value inside a variable cannot be changed after initialization.

Answer:

3. `string city = "Seattle";` stores text data.

Answer:

4. Meaningful variable names improve code readability.

Answer:

5. The following code stores the user's input into a variable.

```
cin>>age;
```

Answer:

Part D. Code Reading (10 pts) Question 1

What is the output?

```
string firstName ="Tom";
string lastName ="Smith";

cout<<firstName+" "+lastName;
```

Answer:

Question 2

What is the output?

```
int score =50;
score =score+20;

cout<<score;
```

Answer:

Part E. Short Answer Questions (10 pts)

1. Explain the difference between a variable **declaration** and a variable **assignment**.

Answer:

2. Why should programmers use meaningful variable names?

Answer:

3. What is string concatenation?

Answer:

Part F. Programming Questions (20 pts)

Question 1 — Personal Introduction Tool

Write a program that asks the user for:

- Name
- Age

Then prints:

```
Hello Alice!
You are 18 years old.
```

Sample Solution

```
#include<iostream>
#include<string>
using namespace std;

int main() {
    string name;
    int age;

    cout<<"Enter your name: ";
    cin>>name;

    cout<<"Enter your age: ";
```

```
cin>>age;

cout<<"Hello "<<name<<"!"<<endl;
cout<<"You are "<<age<<" years old."<<endl;

return 0;
}
```

Question 3 — Daily Study Tracker (Challenge)

Create a tool that asks the user:

```
Hours studied today
Hours studied yesterday
```

Then outputs:

```
Total study time: X hours
```

Example

Input:

```
3
5
```

Output:

```
Total study time: 8 hours
```

Question 4 — Build Your Own Personal Tool (Open-Ended)

Create a tool of your choice that:

- Uses at least **3 variables**
- Uses **user input**
- Uses **string concatenation**
- Produces a personalized output

Examples:

- Fitness Tracker
- Reading Tracker
- Movie Recommendation Tool
- Travel Planner

- Daily Expense Calculator
- Homework Planner

Grading Rubric

Criterion	Points
Correct variable declarations	5
Correct user input handling	5
Meaningful variable names	5
Personalized output	5
Total	20

Bonus Challenge (5 pts)

Predict the output:

```
stringtool ="Study Tracker";  
intdays =7;  
  
cout<<tool<<" used for "  
<<days  
<<" days.";
```

Answer

This set covers all major Lesson 2 concepts: **variables, data types, user input, updating variables, string concatenation, and building a simple personal tool**, while also preparing students for future lessons involving calculations, conditionals, and functions.